# Linear Programming Approach to the Graph Square Problem

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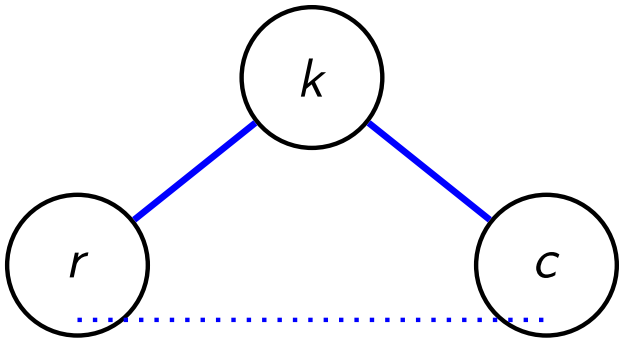
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# Square of a Graph

**Two-step connections**

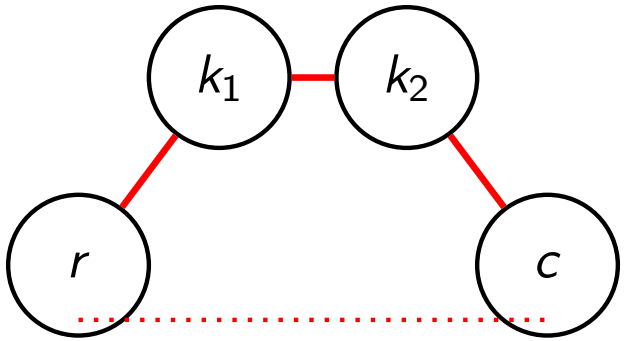



two steps

$$(A^2)_{rc} = \sum_{k=1}^{n} A_{rk} A_{kc}.$$

Counts walks of length two from $r$ to $c$.

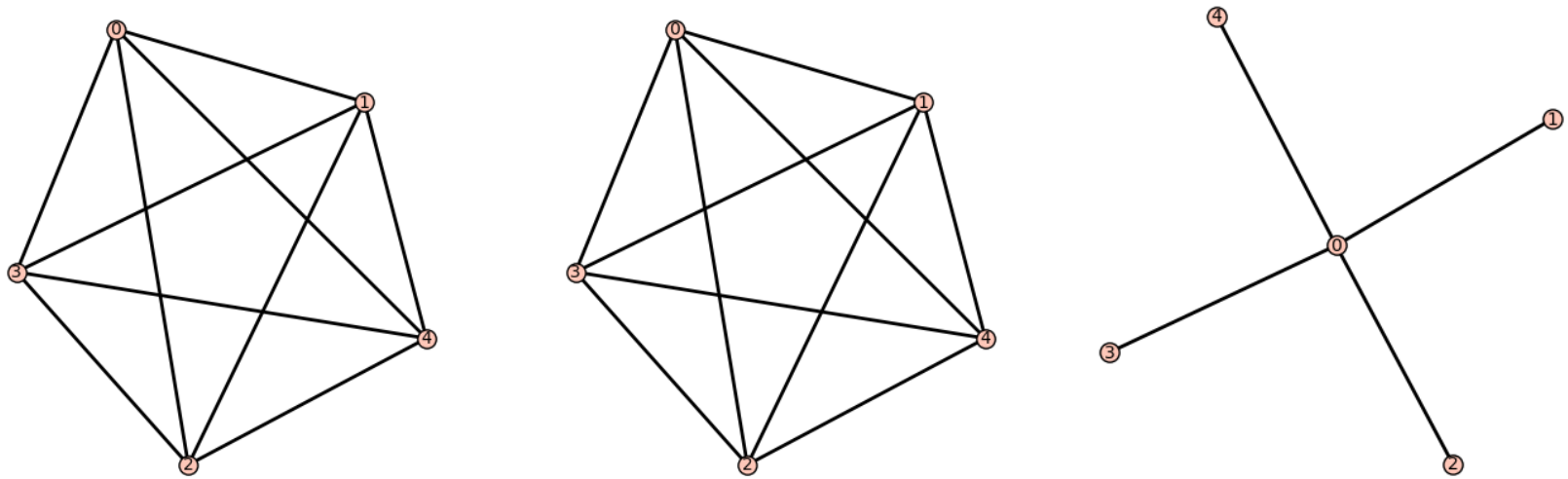
**Three-step connections**




three steps

$$(A^3)_{rc} = \sum_{k=1}^{n} A_{rk} (A^2)_{kc}.$$

Counts walks of length three from $r$ to $c$.




(a) $K_5$. (b) $K_5$, a root of $K_5$. (c) $K_{1,4}$, a root of $K_5$.

Figure: A few of the 15 non-isomorphic graph roots of $K_5$.

# The Graph Square Problem

## Problem (SQUARE)

**Instance:** *A finite, simple, undirected graph* $G = (V, E)$.

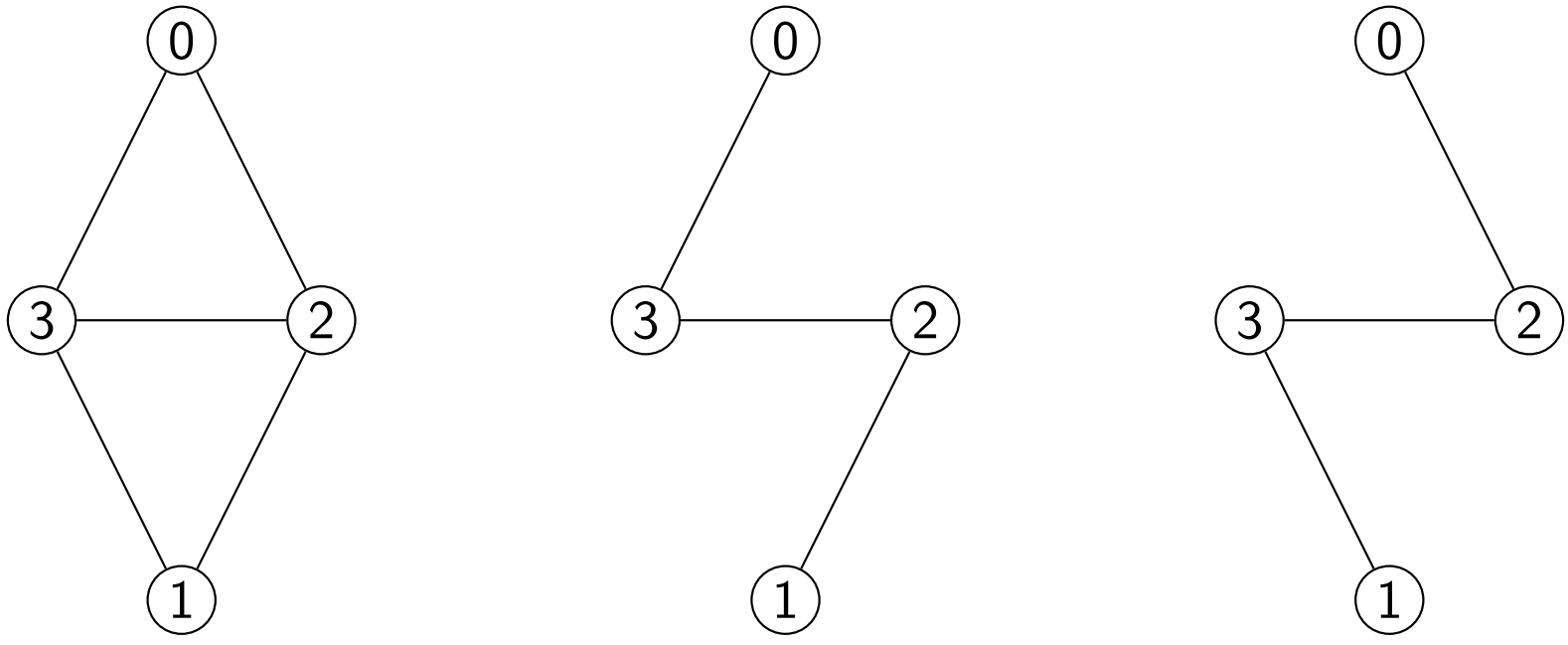
**Question:** *Does there exist a finite, simple, undirected graph* $H$ *such that*

$$G = H^2?$$

Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where $V$ is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. The **square of a graph** $H$, denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \tag{1}$$

That is, two distinct vertices are adjacent in $H^2$ if and only if they are connected by a path of length one or two in $H$. When $G = H^2$ we say that $H$ is a **root** of $G$.




(a) $G$ (b) $T_1$ (c) $T_2$

Figure: The graph $G$ has two possible $P_4$ square roots that are trees, $T_1$ and $T_2$. The two roots are isomorphic, differing only by the interchange of vertices 2 and 3.

# NP-Completeness

Motwani and Sudan (1994) proved that the graph square problem is **NP**-complete by showing that the problem is **NP**-hard through a polynomial reduction from Not-All-Equal 3-SAT.

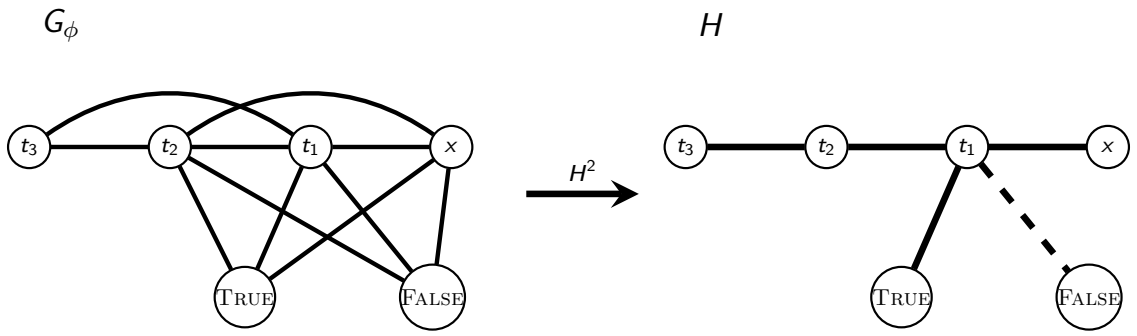
## Problem (NAESAT)

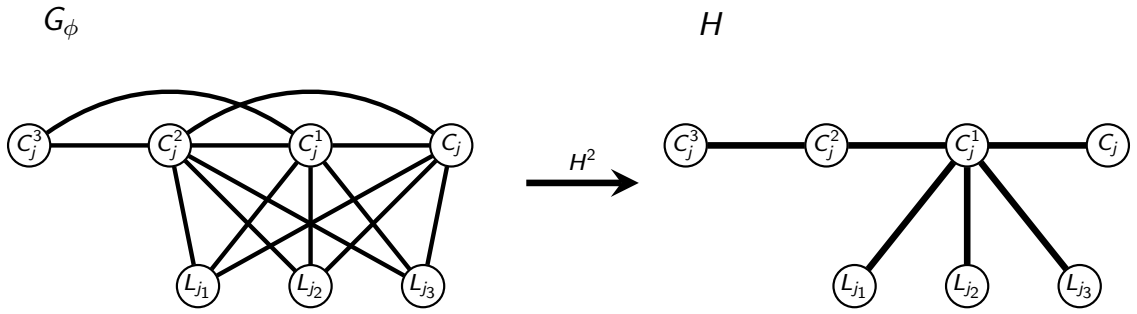
**Instance:** *A 3-CNF formula*

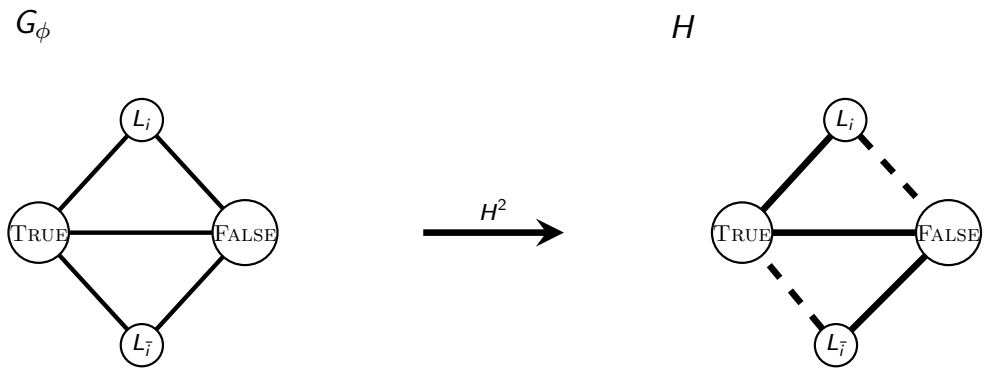
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

*where each clause $C_i$ is over variables $x_1, \ldots, x_n$.*

**Question:** *Does there exist an assignment of* TRUE *and* FALSE *to the variables such that every clause contains at least one* TRUE *literal and at least one* FALSE *literal?*


$G_\phi$
$H$
$t_3$
$t_2$
$t_1$
$x$
TRUE
FALSE
$H^2$


$G_\phi$
$H$
$C_j^3$
$C_j^2$
$C_j^1$
$C_j$
$L_{j_1}$
$L_{j_2}$
$L_{j_3}$
$H^2$


$G_\phi$
$H$
$L_i$
TRUE
FALSE
$L_{\bar{i}}$
$H^2$

# Restricting the Square vs the Root

| **Square $G$** | **Root $H$** | **Main result** |
|---|---|---|
| Unrestricted | Trees | Ross and Harary (1960): structural characterization and reconstruction of tree roots, unique up to isomorphism [RH60]. |
| Unrestricted | Trees | Lin and Skiena (1995): optimal linear-time recognition and reconstruction of tree roots [LS95]. |
| 2-connected | Unrestricted | Fleischner (1974): the square of a 2-connected graph has a Hamiltonian cycle [Fle74]. |
| Interval squares | Unrestricted | Lau and Corneil (2004): polynomial-time construction of roots from interval squares [LC04]. |
| Unrestricted | Block graphs | Le and Tuy (2007): linear-time recognition of roots of block graphs [LT07]. |
| Unrestricted | Bipartite | Lau (2006): linear-time construction of bipartite roots [Lau06]. |
| Directed graphs | Directed graphs | Geller (1970): necessary and sufficient conditions for directed square-root existence [Gel70]. |
| Graphs of prescribed girth | Unrestricted | Farzad, Lau, Le, and Tuy (2009) and Farzad and Karimi (2012): complexity dichotomy for square-root recognition [Far+09; FK12]. |
| Maximum degree $\leq 6$ | Unrestricted | Cochefert, Couturier, Golovach, Kratsch, and Paulusma (2013): minimum-edge roots in polynomial time and maximal-edge roots in exponential time [Coc+13]. |
| Unrestricted | Oriented graphs, maximum degree $\leq 3$, girth $\geq 4$ | Duffy (2025): classification of squares of oriented graphs under these root restrictions [Duf25]. |

# Maximal Cliques and Unicliqual Vertices in Tree Roots

**Input:** A graph $G$ where $G = T^2$
**Output:** A tree $T$
**Step 1**
**if** $G = K_p$ is complete **then**
    **Return** tree $T$ with depth 1, root any vertex
**else**
    **Continue**
**end if**
**Step 2**
Identify all maximal cliques $C_1, \ldots, C_n$ of $G$ (e.g. Bron–Kerbosch [**bron˙kerbosch˙1973˙maximal˙cliques**])
**Step 3**
Let $b_1, \ldots, b_n$ be multicliqual points for $C_1, \ldots, C_n$
Build graph $T$ on $\{b_1, \ldots, b_n\}$
**for** each pair $(C_i, C_j)$ with $|C_i \cap C_j| = 2$ **do**
    Add edge $(b_i, b_j)$ in $T$
**end for**
$T$ is a tree
**Step 4**
**for** each clique $C_i$ **do**
    Let $n_i$ be the number of unicliqual points in $C_i$
    Attach $n_i$ leaves to $b_i$ in $T$
**end for**
Resulting $T$ has $|V(G)|$ nodes and $T^2 = G$

# Integer Program

$$A_G = (A_H)^2 \vee A_H. \tag{2}$$

$$(A_H)^2_{ij} = \big((A_H)_{i1} \wedge (A_H)_{1j}\big) \vee \big((A_H)_{i2} \wedge (A_H)_{2j}\big) \vee \cdots \vee \big((A_H)_{in} \wedge (A_H)_{nj}\big)$$

$$g_{ij} \text{ if and only if } \Big(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij}\Big). \tag{3}$$

## Lemma

*Let $G$ and $H$ be graphs with adjacency matrices $A_G$ and $A_H$,respectively. The graph $H$ is a root of $G$ if and only if*

$$A_G = \mathbb{1}_{\geq 1}\big((A_H)^2 - D_H + A_H\big).$$

# Linear Relaxation

# Min and Max Objective Function

# Identifying Square Characteristics of Tree Roots

# Leaf Fixing Linear Program

# Propagation of the Fixing

# Vertex Deletion

# Symbolic Sets

# Rounding